Chapter 3 exercises

Work over $\mathbb{R}$ unless stated otherwise. Give ambient spaces, bases, dimensions, and conditions where requested, rather than only ranks.

CH03-E1

Determine which sets are subspaces of $\mathbb{R}^2$. Justify each decision using the subspace conditions.

1. $S_1 = \{(x, y)^T : y = 2x\}$.
2. $S_2 = \{(x, y)^T : x = 0 \text{ or } y = 0\}$.
3. $S_3 = \{(x, y)^T : x^2 + y^2 = 1\}$.

CH03-E2

Let

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix}.$$

1. Find RREF, rank, pivot columns, and free columns.
2. Find a basis of $N(A)$.
3. Prove the basis vectors are independent and verify that each satisfies $Ax = 0$.

CH03-E3

For the matrix in CH03-E2 and $b = (2, 5, 3)^T$, solve $Ax = b$.

1. Test consistency using the augmented matrix.
2. Find a particular solution x_p with every free variable zero.
3. Give the complete solution set and decide whether it is a subspace.

CH03-E4

For the matrix of CH03-E2,

1. Find a column-space basis from original pivot columns.
2. Find a row-space basis from nonzero RREF rows.
3. Find a basis of $N(A^T)$.
4. Check $r + (n - r) = n$ and $r + (m - r) = m$, identifying the ambient space for each pair.

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CH03-E5

In the vector space of all real functions $\mathbb{R} \to \mathbb{R}$, consider

$$v_1(x) = 1, \quad v_2(x) = \cos(2x), \quad v_3(x) = \sin^2 x.$$

1. Determine whether the functions are independent on the whole domain.
2. If dependent, give a nontrivial relation.
3. Find the dimension of $\operatorname{span}\{v_1, v_2, v_3\}$.

CH03-E6

Suppose $A \in \mathbb{R}^{3\times 5}$ has rank 2.

1. Give the dimensions and ambient spaces of $C(A), C(A^T), N(A), N(A^T)$.
2. For $b \in C(A)$, how many free parameters describe the solution set?
3. Does an inconsistent right-hand side exist? Explain.
4. Is the added assumption $\dim N(A) = 0$ compatible with the stated rank?

CH03-E7

Let V have basis $\{u_1, u_2, u_3\}$, and let $w_1, w_2, w_3 \in V$ satisfy

$$[w_1\ w_2\ w_3] = [u_1\ u_2\ u_3]B, \qquad B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

1. Use the rank of B to test independence without manipulating the abstract vectors directly.
2. Find the dimension of their span.
3. Give and verify a nontrivial dependence relation.

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CH03-E8

Using CH03-E2 and CH03-E4, take

$$x = (-2, 3, 0, 3)^T.$$

1. Use bases of $C(A^T)$ and $N(A)$ to find $x = x_{\text{row}} + x_{\text{null}}$.
2. Verify membership of each component in its stated space and that their sum is x.
3. Prove independence of the four basis vectors and explain why it makes the decomposition unique.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 3.

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